

OPENAGENCY: SELF-EVOLVING MULTI-AGENT SYSTEM VIA GOVERNED COLLABORATION PROTOCOLS

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ABSTRACT

Large language model (LLM) agents have shown strong capabilities on complex tasks, encouraging multi-agent systems that coordinate specialized roles. Improving how these agents collaborate requires flexibility, so that coordination decisions are still made at runtime, together with controllability, so that they can be inspected and revised afterwards. Existing systems supply only one of the two, because a predefined workflow fixes the execution route while free-form communication leaves the realized coordination opaque. In this paper, we introduce OpenAgency, a self-evolving multi-agent system built on governed collaboration protocols. A protocol defines roles, admissible handoffs, handoff modes, and completion conditions, and each agent still selects its next collaborator inside the admissible set, which turns the collaboration pattern itself into an evolvable object. To evolve that object, OpenAgency keeps an immutable trace that retains rejected handoffs and failed completions alongside successful events, and a collaboration-driven evolving agent converts this evidence into a bounded number of atomic protocol edits, each screened by a structural validity operator and by the collaboration quality the agent scores from the realized coordination. Experiments on fourteen benchmarks spanning general question answering, long-horizon tasks, and production-oriented workloads show that OpenAgency reaches the highest primary metric among the compared systems on every benchmark.

1 INTRODUCTION

Large language models (LLMs) (OpenAI, 2023; 2022) have shown strong capabilities on complex tasks such as software engineering (Yang et al., 2024b; Wang et al., 2024b; Zhang et al., 2024a), data research (Liu et al., 2026; Li et al., 2024), and long-horizon planning (Zhang et al., 2026; Xu et al., 2024). Multi-agent systems built on LLMs extend these capabilities by assigning complementary roles that exchange intermediate evidence (Wang et al., 2024a; Xi et al., 2023; Hong et al., 2024; Qian et al., 2024; Wu et al., 2023), which lets one system plan and verify the same task through cooperation. However, as tasks become longer and more heterogeneous, the collaboration process itself becomes both a source of errors and an underused source of improvement.

Evolving the way agents collaborate requires flexibility and controllability at the same time. Flexibility matters because a multi-agent system has to recover what went wrong across roles, and one that cannot repair a failed handoff at runtime loses the remainder of the run with it. Controllability is equally necessary, since an update is informative only when the system can tell which coordination step was responsible and can verify that the revised collaboration remains admissible. Once either property is missing, cross-run improvement falls back to a blind search driven by the final score. For instance, on SWE-bench Verified (Jimenez et al., 2024), a free-form multi-agent baseline reaches only 51.4% file match and a standard operating procedure (SOP) baseline reaches 59.8%, far below the 86.2% attained once the collaboration itself can be revised from execution traces (Table 3).

Existing collaboration representations hold only one of these two properties. *Blueprint-driven* systems encode a graph, workflow, or standard operating procedure (Hong et al., 2024; Wu et al., 2023; Fournery et al., 2024), and recent methods automate this design through workflow search or generated agent programs (Zhang et al., 2025b; Hu et al., 2025; Zhang et al., 2025a). Such designs are controllable, yet each selected workflow still determines the exact route taken during execution, and a runtime deviation therefore cannot be repaired locally. *Agent-driven* systems let debate, dialogue,

or dynamic connectivity emerge at runtime (Li et al., 2023; Du et al., 2023; Liu et al., 2024; Park et al., 2023), and a learned orchestrator can further reorder agent activations from reward (Dang et al., 2025). Flexibility here costs transparency, since the realized coordination stays opaque and is hard to attribute or revise across runs. One line of work fixes the route and the other hides it, so neither yields a representation that is checkable at runtime and revisable from experience.

In this paper, we aim to make the collaboration pattern itself evolvable by holding flexibility and controllability in one representation. To this end, we introduce *OpenAgency*, a self-evolving multi-agent system built on *governed collaboration protocols*. As illustrated in Figure 1, OpenAgency represents a protocol as a tuple $\mathcal{P} = (\mathcal{R}, \mathcal{N}, \alpha, \mathcal{H}, \mu, \Omega, \mathcal{C})$ whose components fix the roles and the admissible handoffs between them. The admissible handoff set $\Gamma(n, s)$ carries both properties in a single object, since it rules out inadmissible transitions while the choice inside it is left to the agent at runtime. Because every coordination decision is expressed in this object and its legality is decidable, the collaboration pattern becomes revisable between runs. Building on this, OpenAgency records accepted handoffs, rejected handoffs, completion events, and failures in an immutable trace, and a collaboration-driven evolving agent turns this evidence into at most k atomic edits over the protocol. A validity operator Π_V filters malformed topologies and the same agent scores the coordination each surviving candidate realizes, so every deployed round stays admissible and traceable to the events that motivated it.

Experiments across fourteen benchmarks covering general question answering, long-horizon tasks, and production-oriented tasks show that OpenAgency achieves the best primary metric on every benchmark, lifting the unstructured multi-agent initialization by 20.7 absolute points on average (Table 1). To the best of our knowledge, OpenAgency is the first multi-agent system in which the collaboration pattern is an evolvable object, updated from immutable execution traces without regenerating the whole workflow. We summarize our contributions as follows.

- **Evolvable Collaboration Protocol:** We make the collaboration pattern itself a governed and evolvable object, because the admissible handoff set is checkable before a transition is taken and revisable after the run that produced it.
- **Collaboration-driven Evolution:** We propose an evolving agent that learns from the coordination evidence prior systems discard, namely rejected handoffs and failed completions, and converts it into at most k atomic edits over protocol structure, predicates, and shared context. A validity operator and a collaboration quality screen keep every deployed round admissible and free of regression.
- **Strong Performance:** OpenAgency reaches the best primary metric on all fourteen benchmarks (Section 4), and the same advantage holds against free-form and SOP collaboration substrates under an identical executor. Ablations further show that the freedom to change the agent count is the level that pays most, and a fixed graph saturates thirty points below the full space once one structural edit is admitted.

2 RELATED WORK

Multi-agent Collaboration. Existing systems expose collaboration with different degrees of structure. Blueprint-driven approaches encode graphs, workflows, or standard operating procedures (Hong et al., 2024; Qian et al., 2024; Wu et al., 2023). These systems make coordination explicit but typically fix the selected topology before execution. Agent-driven systems allow collaboration to emerge from debate (Du et al., 2023; Liang et al., 2023), role-conditioned dialogue (Li et al., 2023; Chan et al., 2023), or dynamic connectivity (Liu et al., 2024; Zhang et al., 2024b). Collaboration then adapts during execution, though the realized structure remains difficult to inspect and replay. Recent methods also optimize communication for accuracy or efficiency (Sun et al., 2023; Chen et al., 2024a;b). Across both families, the collaboration pattern is treated as a fixed design choice or as an emergent byproduct, and neither case leaves an object that later runs can update.

Agent Self-evolution. Agent systems can improve through reasoning feedback, memory, prompt optimization, or program search. Reflexion and Self-Refine use rollout feedback to revise subsequent reasoning (Shinn et al., 2023; Madaan et al., 2023), while AgentGym and experiential co-learning accumulate experience across interactions (Xi et al., 2024; Qian et al., 2023). DSPy, TextGrad, and OPRO optimize language-model programs or prompt strings (Khattab et al., 2024; Yuksekogonul et al., 2024; Yang et al., 2024a). More recently, AFlow, automated agent design, and

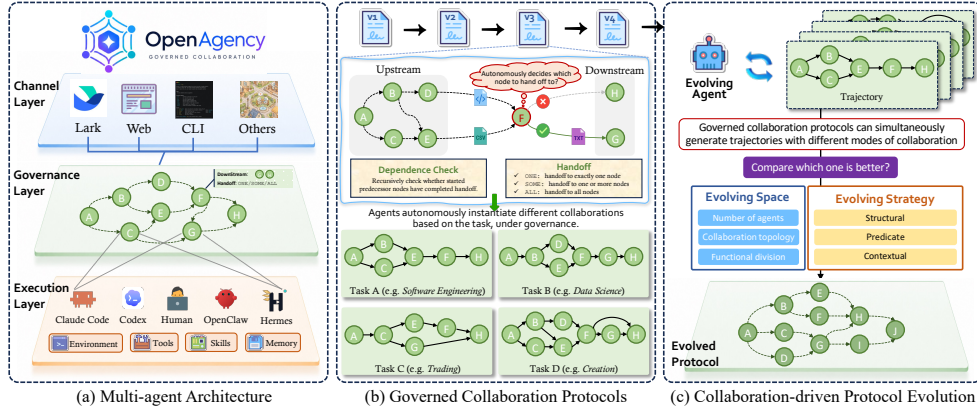


Figure 1: Overview of OpenAgency. (a) Three layers keep the collaboration protocol independent of the backend that realizes each role. (b) Under one governed protocol the agents choose their own handoffs, so different tasks instantiate different collaborations. (c) The evolving agent reads the collected trajectories and returns the next protocol, moving inside the evolving space under one of three edit strategies.

MaaS search over workflow graphs or generated agent programs (Zhang et al., 2025b; Hu et al., 2025; Zhang et al., 2025a), while evolving orchestration trains a central controller to reorder agent activations (Dang et al., 2025). These methods show that collaboration structure can be optimized, but what they return is an executable topology or a controller policy, neither of which exposes a boundary that a later round can check a candidate revision against.

The open question is therefore which representation of collaboration can itself be evolved. OpenAgency defines a governed protocol that fixes admissible handoffs and completion conditions without committing to one route, so agents keep local choices during execution while every decision remains checkable, which is what makes the protocol revisable from immutable traces across runs. OpenAgency therefore evolves the governance boundary that admits runtime decisions, while workflow search and orchestrator learning optimize the executed route or the policy that produces it.

3 OPENAGENCY

OpenAgency treats collaboration as a first-class object separated from the executors that realize each role. It combines a layered architecture (Section 3.1), a governed protocol that makes coordination inspectable (Section 3.2), and bounded protocol evolution driven by execution traces (Section 3.3), as Figure 1 illustrates.

3.1 MULTI-AGENT ARCHITECTURE

To apply one collaboration protocol across applications with heterogeneous executors, OpenAgency separates multi-agent collaboration into three layers that keep the protocol independent of backend-specific prompts and actions.

Channel Layer. The channel layer exposes the collaboration to its users through an enterprise chat workspace, a web console, or a command line, and keeps instructions, agent messages, human feedback, and coordination events on one shared timeline.

Governance Layer. The governance layer hosts the collaboration protocol and validates each proposed handoff against the current protocol state. It records both accepted and rejected events in an immutable trace that supplies the evidence for later protocol evolution.

Execution Layer. The execution layer binds each role to a concrete executor. An executor can be an LLM agent, a coding harness, a domain-specific tool, or a human reviewer, and it sits below the governance layer, so the protocol constrains how roles hand off whichever backend realizes a role.

We formalize a running OpenAgency instance at step t as $\mathcal{I}^{(t)} = (\mathcal{X}^{(t)}, \mathcal{P}, \mathcal{E}, s^{(t)})$, where $\mathcal{X}^{(t)}$ is the shared context, $\mathcal{P}$ is the governed protocol, $\mathcal{E}$ is the set of available executors, and $s^{(t)}$ is the current protocol state. The map $\rho : \mathcal{R} \rightarrow \mathcal{E}$ binds each role in the role set $\mathcal{R}$ to an executor and stays separate from the protocol logic. The governance layer renders the shared state into a role-specific execution context $u^{(t)} = f_{\text{ctx}}(\mathcal{X}^{(t)}, s^{(t)}, n, r, \mathcal{P})$ for the active node n with assigned role r , since an LLM executor may require tool instructions while a human executor may require a concise protocol description. Adapters return messages, handoffs, finishes, and failures in a common event representation, so the executor bound to a role can change without changing the handoff rules, trace format, or completion predicates.

3.2 GOVERNED COLLABORATION PROTOCOLS

To govern collaboration without fixing a complete execution route, OpenAgency represents collaboration as a declarative tuple that exposes enough structure to validate each handoff while preserving the agent’s choice among admissible targets.

Protocol Definition. OpenAgency represents a governed collaboration protocol as

$$\mathcal{P} = (\mathcal{R}, \mathcal{N}, \alpha, \mathcal{H}, \mu, \Omega, \mathcal{C}), \quad (1)$$

where $\mathcal{R}$ is the set of roles, $\mathcal{N}$ the set of collaboration nodes, $\alpha : \mathcal{N} \rightarrow \mathcal{R}$ assigns each node to a role, $\mathcal{H} \subseteq \mathcal{N} \times \mathcal{N}$ defines admissible handoff edges, $\mu(n) \in \{\text{ONE}, \text{SOME}, \text{ALL}\}$ is the handoff mode, Ω_n is the completion predicate, and $\mathcal{C}$ contains the shared rule and culture that every role sees.

Admissible Handoff Set. For a node n under state s , the admissible handoff set is $\Gamma(n, s) = \{n' \in \mathcal{N} \mid (n, n') \in \mathcal{H} \text{ and the transition is valid under } s\}$. The handoff mode $\mu(n)$ then determines how the active agent selects targets from $\Gamma(n, s)$. Under ONE the agent picks exactly one admissible target, under SOME a nonempty subset, and under ALL every admissible target. A node with no admissible target completes without a handoff, and the local choice within $\Gamma(n, s)$ is left to the agent.

Handoff Validation and Completion. When the agent proposes a handoff (n, B, m) with target set B and message m , the protocol validates B under $\mu(n)$. An admissible handoff commits B as pending targets and updates the state, and an inadmissible one leaves the state unchanged and returns structured feedback so the agent may propose another target. Committing a target set does not activate it. A finish event (n, y) confirms that the source node has completed its work, and the governance layer activates the committed targets only when $\Omega_n(y)$ holds, so downstream execution waits for the completion predicate even after an admissible target set is committed. Activation runs one further dependence check, which walks back from each committed target and confirms that every started predecessor of that target has already handed off, so a node begins only once the collaborators it depends on have delivered.

Algorithm 1 states the resulting runtime interpreter. The state advances only on admissible handoffs and valid completions, while the trace keeps every event, including the ones the boundary refused, so later evolution reads both successful and unsuccessful coordination.

3.3 COLLABORATION-DRIVEN PROTOCOL EVOLUTION

Given a protocol whose decisions are explicit and checkable, the remaining question is how to revise it from experience. Revising a protocol takes the shape of an optimization problem, since it needs a signal that says where the current object is wrong, a space of admissible moves, and a rule that decides which move to keep. The final answer alone cannot indicate which of a run’s many coordination events should change, and regenerating the whole workflow each round would destroy the governance boundary that makes a change inspectable. A single evolving agent therefore drives the loop. It reads a search direction off the collaboration trace, moves inside a bounded space of protocol edits, and keeps a move only when that move survives an admissibility and quality screen.

Trace Diagnosis as Search Signal. Each execution produces an ordered trace $\tau = (e_1, s_1), \dots, (e_T, s_T)$ of coordination events with the resulting states, where an event e_t is one of accepted handoff, rejected handoff, finish, message, or failure. Unlike prior systems that keep only the successful trajectory or the final score, the trace retains rejected handoffs and failed completions,

Algorithm 1 Collaboration under a Governed Protocol

Input: Protocol $\mathcal{P}$, initial context $\mathcal{X}$, executor mapping $\rho: \mathcal{R} \rightarrow \mathcal{E}$
Output: Immutable collaboration trace τ

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1:  $\tau \leftarrow \emptyset$ ; activate start node  $n_0$ ; obtain initial state  $s$ 
2: while active nodes exist do
3:   for each active node  $n$  with role  $r = \alpha(n)$  do
4:      $e \leftarrow \rho(r)(f_{\text{ctx}}(\mathcal{X}, s, n, r, \mathcal{P}))$  ▷ executor proposes an event
5:     if  $e$  is a handoff  $(n, B, m)$  and  $B$  is admissible then
6:       commit  $B$  as pending targets and update  $s$ 
7:     else if  $e$  is a finish  $(n, y)$  and  $\Omega_n(y)$  holds then
8:       mark  $n$  complete, then activate every committed target whose started predecessors have handed
      off
9:     else
10:      return structured feedback to  $\rho(r)$  ▷ refused at the governance boundary
11:      append  $(e, s)$  to  $\tau$  ▷ accepted and refused events are both retained
12: return  $\tau$ 

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which record the transitions the collaboration attempted and could not take. The evolving agent runs an analyzer f_{diag} over $\tau^{(1:v)}$ and returns the coordination sites that account for the observed failures, namely the handoff edges the agents repeatedly attempted and the admissible set refused, together with the completion predicates that fired without a usable output. This signal plays the role a gradient plays in parameter optimization, since it attributes a run-level outcome to individual coordination decisions and points at the part of the protocol a revision should touch.

Factorized Evolving Space. The search space factorizes into three levels. *Number of agents* covers the node count $|\mathcal{N}|$. *Collaboration topology* covers the handoff edge set $\mathcal{H}$ and the handoff modes μ . *Functional division* covers the role assignment α , the completion predicates Ω , and the shared rule and culture in $\mathcal{C}$, which together fix what each node owes the collaborators downstream of it. A point in this space is not one execution route, because each agent still chooses inside $\Gamma(n, s)$, so a single protocol unfolds into many realized collaboration patterns at inference time, as the four task instantiations of Figure 1 show. The optimizer therefore searches far fewer points than the collaborations they realize. The factorization also makes comparison readable, since a candidate produced by a single-level edit shares the remaining levels with the incumbent, so an overloaded role can be told apart from a missing edge.

Bounded Protocol Optimizer. A move in this space is a composite edit of at most k atomic changes drawn from three edit strategies. A *structural* edit adds or removes a node or a handoff edge, a *predicate* edit rewrites a completion or transition condition, and a *contextual* edit rewrites the rule and culture a role reads. At round v the optimizer f_{evolve} takes one such step from the incumbent under the analyzed trace signal:

$$\hat{\mathcal{P}}^{(v+1)} = f_{\text{evolve}}(\mathcal{P}^{(v)}, f_{\text{diag}}(\tau^{(1:v)})), \quad |\hat{\mathcal{P}}^{(v+1)} \ominus \mathcal{P}^{(v)}| \leq k. \quad (2)$$

We set $k = 4$, which bounds the step the way a learning rate bounds a parameter update and keeps each candidate a small inspectable change. The strategy is a bounded greedy ascent, proposing one candidate per round and retaining the incumbent whenever the candidate fails the screen below, so the deployed sequence never moves downhill in collaboration quality.

Admissibility and Quality Screen. A step is screened twice before it is deployed. The first check is a structural validity operator $\Pi_{\mathcal{V}}$ on the candidate topology, which rejects malformed handoff edges, unresolved role references, duplicate node identifiers, and completion states no path can reach, so a candidate that is not a well-formed protocol never runs. Writing $\tilde{\mathcal{P}}^{(v+1)} = \Pi_{\mathcal{V}}(\hat{\mathcal{P}}^{(v+1)})$ for the validity-checked candidate, the second check executes it and hands the resulting traces to the evolving agent. The agent returns a collaboration quality score q_{col} from the realized coordination, reading whether each role delivered what its successors required and whether the completion conditions closed on a usable output. The deployment rule is

$$\mathcal{P}^{(v+1)} = \begin{cases} \tilde{\mathcal{P}}^{(v+1)}, & \tilde{\mathcal{P}}^{(v+1)} \neq \perp \wedge q_{\text{col}}(\tilde{\mathcal{P}}^{(v+1)}) \geq q_{\text{col}}(\mathcal{P}^{(v)}), \\ \mathcal{P}^{(v)}, & \text{otherwise,} \end{cases} \quad (3)$$

so a candidate is deployed only when it is well formed and does not degrade the observed coordination. Both checks read the candidate protocol and the traces it produces, which keeps the update loop closed over collaboration evidence.

4 EXPERIMENTS

We evaluate the governed protocol of Section 3 on fourteen benchmarks (Section 4.1) under one executor and tool budget (Section 4.2), and report the main comparison in Section 4.3. Section 5 then locates the gain, through protocol evolution round by round (Section 5.1), its component measurements (Section 5.2), the levels of the evolving space (Section 5.3), multi-agent protocols (Section 5.4), free-form and SOP substrates (Section 5.5), and a fixed collaboration graph (Section 5.6). Appendix C reports the activity of the governance boundary.

4.1 BENCHMARKS

We evaluate OpenAgency on fourteen benchmarks organized into three categories. *General question answering* contains HumanEval (Chen et al., 2021), MBPP (Austin et al., 2021), GSM8K (Cobbe et al., 2021), MATH (Hendrycks et al., 2021), and HotpotQA (Yang et al., 2018), and stresses whether protocol adaptation helps when each instance admits a short answer. *Long-horizon tasks* contains SWE-bench Verified (Jimenez et al., 2024), DeepPlanning (Zhang et al., 2026), and the text-only subset of Humanity’s Last Exam (HLE) (Phan et al., 2025), and stresses whether governed handoffs support dependent decisions. *Production-oriented tasks* contains DevEval (Li et al., 2024), DDR-Bench (Liu et al., 2026), TheAgentCompany (Xu et al., 2024), MARBLE (Zhu et al., 2025), AutomationBench (Shepard & Salimans, 2026), and OfficeQA Pro (Opsahl-Ong et al., 2026), and stresses whether one protocol representation applies across heterogeneous tools and task surfaces. Appendix A describes each benchmark.

4.2 EXPERIMENTAL SETUP

To separate governed collaboration from the executor’s own capability, we use GPT-5.5 across all benchmarks, and each role receives the tool surface its benchmark requires. For each benchmark, we compare three in-house conditions under one model and tool budget, a single-agent ablation with no handoff, the unstructured multi-agent initialization v_0 , and the evolved protocol after four rounds (v_4). All three are reported against a prior reference (Prior Ref.) that fixes the absolute level of the benchmark, taken as the strongest publicly reported result we found for each benchmark at the time of submission, and as a same-model direct-LLM baseline where no stronger external result is available. Each benchmark evolves its own protocol independently, and the evolving agent reads only execution traces, so no task label or reference answer reaches it, following the label-free self-refinement setting of Shinn et al. (2023) and Madaan et al. (2023). Model, temperature, and tool budget stay fixed across rounds, and only the protocol template changes. Table 1 reports the final round and Appendix D lists all four.

4.3 MAIN RESULTS

Capability on General Question Answering. The five single-turn benchmarks (HumanEval, MBPP, GSM8K, MATH, HotpotQA) probe *short-chain reasoning capabilities*, since each instance has a localized, automatically verifiable output and leaves little room to coordinate. Against its two ablations and each benchmark’s prior reference, OpenAgency obtains the best primary metric on all five in Table 1, reaching 100.0 pass@1 on HumanEval and lifting MBPP from 73.5 to 80.9 over the initialization. Agent count alone does not produce this gain, since the unstructured initialization trails the single-agent ablation on HotpotQA at 64.0 against 75.9 F1 while the governed protocol carries the same benchmark to 82.3. Evolution prunes handoffs whose upstream output the trace flags as ineffective, and each benchmark settles on a solve-critique-revise loop matched to its dominant failure mode, so the protocol turns a short chain into a verified one.

Capability on Long-horizon Tasks. SWE-bench Verified, DeepPlanning, and HLE stress *long-horizon reasoning capabilities*, since each instance requires several dependent decisions and a bad handoff at any stage can invalidate the entire run. Against the single-agent ablation and each benchmark’s prior reference, Table 1 reports a lead on all three, reaching 86.2 file match on SWE-bench Verified against a prior reference of 42.8 and 37.7 accuracy on HLE. The protocol each benchmark converges to shows what produces the lead. On SWE-bench Verified the evolving agent installs a retry edge when a patch fails and splits an overloaded Verifier into structural and semantic components. The DeepPlanning protocol hands a Verifier the user-specified constraints one at a time,

Table 1: OpenAgency and its ablations against a selected prior reference that anchors the absolute level of each benchmark. Multi-agent (init) is the unstructured initialization and OpenAgency is the evolved protocol, both sharing the GPT-5.5 executor and tool budget of the single-agent ablation. Each row uses the benchmark’s native metric, where EM is exact match, FM is file match, and AT is the acceptance-test pass rate.

Benchmark	Metric	Prior Ref.	Single-agent	Multi-agent (init)	OpenAgency
<i>General Question Answering Tasks</i>					
HumanEval	pass@1	92.2	87.5	96.9	100.0
MBPP	pass@1	69.3	73.5	73.5	80.9
GSM8K	EM	97.6	97.2	97.4	98.0
MATH	EM	83.3	82.3	86.0	87.0
HotpotQA	F1	76.4	75.9	64.0	82.3
<i>Long-horizon Tasks</i>					
SWE-bench Verified	FM	42.8	75.6	67.0	86.2
DeepPlanning	Case Acc	46.9	45.0	33.8	54.2
HLE (text-only)	Accuracy	—	18.4	5.6	37.7
<i>Production-oriented Tasks</i>					
DevEval	AT	62.5	62.5	82.1	96.9
DDR-Bench	Overall	69.1	42.8	44.0	76.0
TheAgentCompany	Score	41.5	46.3	41.5	53.7
MARBLE	Overall	79.3	78.4	50.0	98.9
AutomationBench	Acc	42.5	23.7	23.6	54.8
OfficeQA Pro	Acc@1%	57.1	35.3	21.1	69.9

turning a brittle end-to-end plan into locally verifiable handoffs. Verification on HLE becomes conditional, routing high-confidence answers to completion and low-confidence ones back through a retry edge. All three shapes come from the same initialization, so the handoff and completion rules adapt when local errors are unrecoverable.

Capability on Production-oriented Tasks. The six production-oriented benchmarks (DevEval, DDR-Bench, TheAgentCompany, MARBLE, AutomationBench, OfficeQA Pro) test *cross-domain deployment capabilities*, since each exposes a different tool surface and task format. OpenAgency leads on all six in Table 1, improving over the prior reference by 34.4 points on DevEval, 19.6 points on MARBLE, and 12.8 points on OfficeQA Pro. The template differs per benchmark, from a Reader-Solver-Verifier chain on OfficeQA Pro to a Generator-Critic-Reviser pipeline on MARBLE, and two cases expose what governance contributes. On DDR-Bench, an Auditor role introduced during evolution forces every quantitative finding in the trace into the final report, fixing a silent omission that free-form baselines leave undetected. The admissible-set constraint on MARBLE keeps each handoff terminating under a protocol that would otherwise allow any-to-any messaging, so in both cases governance supplies a constraint the task format alone does not.

5 ANALYSIS

5.1 EFFECTIVENESS OF PROTOCOL EVOLUTION

To verify that bounded edits keep improving collaboration after the first revision, we track the primary metric from the initialization v_0 to the evolved rounds v_1 through v_4 on all fourteen benchmarks. Figure 2 and Table 5 record an improvement over the initialization on every benchmark, reaching 48.9 points on MARBLE and 48.8 points on OfficeQA Pro. The deployed sequence is non-decreasing on every benchmark in collaboration quality, since the screen keeps the incumbent whenever a candidate scores lower. On the fourteen benchmarks studied, this ordering also holds for the primary task metric.

5.2 COMPONENT MEASUREMENTS BEHIND THE GAIN

To locate the gain inside a benchmark, we decompose the SWE-bench Verified and DevEval results of every round into their component measurements. Table 2 shows the two improving at different points. On SWE-bench Verified the file-match rate given a patch moves only from 84.6 to 86.2, while

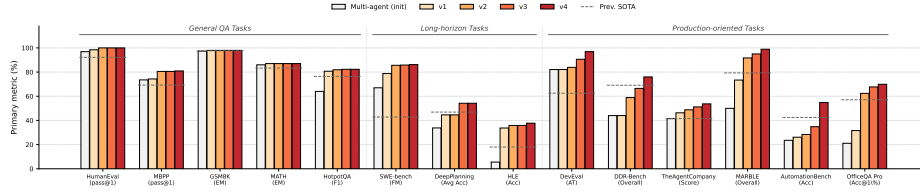


Figure 2: Round-wise trajectory of the primary metric from the initialization v_0 through the evolved rounds v_1-v_4 . Grey dashed lines mark each benchmark’s prior reference.

Table 2: Component measurements across the evolution rounds. Patch applied is the fraction of instances producing a non-empty patch, and file match given a patch restricts file match to those instances.

Component measurement	Init	v_1	v_2	v_3	v_4
<i>SWE-bench Verified</i>					
Patch applied	79.2	93.4	100.0	100.0	100.0
File match	67.0	78.8	85.6	85.8	86.2
File match given a patch	84.6	84.4	85.6	85.8	86.2
<i>DevEval</i>					
Unit test	98.1	98.1	97.3	99.1	99.1
Acceptance test	82.1	82.1	83.9	90.6	96.9

the fraction of instances producing a patch at all rises from 79.2 to 100.0, so the gain is consistent with closing the no-output failure mode that the retry edges of Section 3.3 target. On DevEval the unit-test rate stays near saturation while acceptance rises from 82.1 to 96.9, so the admitted edits are associated with the repository-level correctness that unit tests leave unchecked. In both cases the component that moves aligns with the edit the evolving agent targeted.

5.3 ABLATION OF THE EVOLVING SPACE

To identify which levels of the evolving space carry the gain, we restrict that space in two nested steps and rerun the same evolution loop on three benchmarks. The full arm edits all three levels of Section 3.3. Holding the number of agents $|\mathcal{N}|$ at the initialization gives the fixed-count arm, which still edits the handoff structure $(\mathcal{H}, \mu)$ and the functional division, and freezing the whole graph gives the division-only arm, which reaches only the role assignment α , the completion predicates Ω , and the shared $\mathcal{C}$. Figure 3 reports the best protocol each arm reaches, and the three benchmarks order the arms identically. Freezing the agent count alone costs 5.8 points on SWE-bench Verified, 12.6 on DevEval, and 15.3 on DDR-Bench, so on these three benchmarks the freedom to change the agent count is the level that pays most, and freezing the remaining handoff structure costs a further 2.3 to 5.2 points. A hand-designed system commits both structural levels before any trace has been read, while one governed protocol defers that choice to the trace.

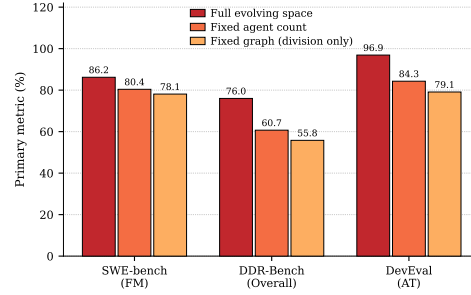


Figure 3: Three nested evolving spaces under one initialization and one evolution loop. The fixed-count arm holds the number of agents $|\mathcal{N}|$, and the division-only arm freezes the whole graph and edits only the functional division.

5.4 ADVANTAGE OF MULTI-AGENT PROTOCOLS

To attribute the gain to governance and not to agent count, we compare OpenAgency with its single-agent ablation under one GPT-5.5 executor, identical tool access, and identical temperature. OpenAgency exceeds that ablation on every benchmark in Figure 4, with the widest gaps on OfficeQA Pro, DevEval, and DDR-Bench. A single node interleaves solving and checking inside one context, while the governed protocol routes a partial output to a downstream role whose completion predicate

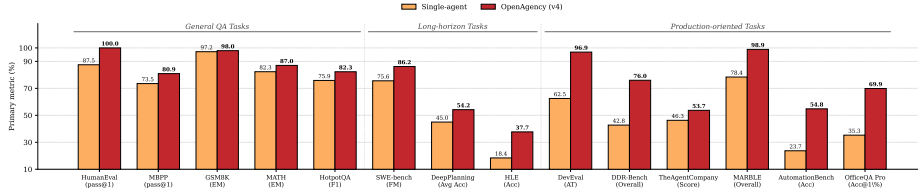


Figure 4: OpenAgency versus its single-agent ablation under identical model and tool budget, isolating governed collaboration from raw solver capacity.

Table 3: OpenAgency against the free-form and SOP baselines.

System	SWE-bench	DevEval	DDR-Bench	TheAgentCompany
AutoGen	51.4	68.9	39.2	38.7
ChatDev	57.6	78.3	44.1	41.9
MetaGPT	59.8	79.5	46.8	42.3
OpenAgency	86.2	96.9	76.0	53.7

validates it independently. The initialization column of Table 1 places the gain on protocol design, since the same initial agent count without governance falls short.

5.5 COMPARISON AGAINST FREE-FORM AND SOP BASELINES

To compare the three collaboration substrates directly, we evaluate OpenAgency against the free-form baseline AutoGen (Wu et al., 2023) and the SOP baselines ChatDev (Qian et al., 2024) and MetaGPT (Hong et al., 2024) on four benchmarks. OpenAgency obtains the highest score in Table 3, exceeding the strongest SOP baseline by 26.4 points on SWE-bench Verified. The free-form baseline is the weakest everywhere, and the SOP baselines recover part of the gap because the collaboration order is enforced, yet their fixed procedure absorbs no trace-level failure signal, so the remaining gap tracks what stays revisable after deployment.

5.6 SUPERIORITY OF GOVERNED PROTOCOLS OVER FIXED GRAPHS

To measure what a fixed collaboration graph costs round by round, we plot the OfficeQA Pro evolution record as two arms from one initialization, where the previous-protocol arm tunes a prompt over that fixed structure. Figure 5 shows six functional-division rewrites peaking at 31.6 in round 2, with the four later rewrites all falling back below it. Widening the graph moves it at once, since one added reader node reaches 62.4 and one added verifier node reaches 66.2. On OfficeQA Pro, a fixed graph keeps the search inside a level the trace already reports as saturated, while one structural edit clears the previous-protocol ceiling by more than thirty points.

6 CONCLUSION

We introduced OpenAgency, a self-evolving multi-agent system built on governed collaboration protocols. Constraining runtime handoffs while preserving each agent’s choice among admissible collaborators turns the collaboration pattern into an evolvable object, and ordered traces become bounded, validity-screened protocol updates. OpenAgency reaches the highest primary metric among the compared systems on each of fourteen benchmarks.

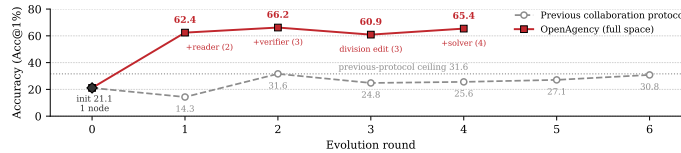


Figure 5: Evolution on OfficeQA Pro from one initialization, scored on the same 133 tasks. The previous-protocol arm edits only the functional division, while the full arm may also change the agent count and the topology.

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A BENCHMARK DETAILS

HumanEval (Chen et al., 2021) evaluates function-level Python code synthesis from natural-language docstrings.

MBPP (Austin et al., 2021) evaluates short-form program synthesis from natural-language descriptions.

GSM8K (Cobbe et al., 2021) evaluates multi-step reasoning over grade-school math word problems.

MATH (Hendrycks et al., 2021) evaluates competition-level mathematical reasoning across five subject areas.

HotpotQA (Yang et al., 2018) evaluates multi-hop question answering that links evidence from multiple documents.

SWE-bench Verified (Jimenez et al., 2024) evaluates repository-level software repair on human-validated GitHub issues, where the agent must produce a patch that resolves the reported bug against a hidden test suite.

DeepPlanning (Zhang et al., 2026) evaluates travel and shopping planning under user-specified constraints, and the primary metric is mean case-level accuracy across the two task families.

Humanity’s Last Exam (HLE) (Phan et al., 2025) evaluates expert-level question answering across mathematics, physics, biology, humanities, computer science, and engineering, and we use its 2,158-question text-only subset.

DevEval (Li et al., 2024) evaluates repository-level code generation under acceptance-test oracles that check whether the generated code works end-to-end with existing modules.

DDR-Bench (Liu et al., 2026) evaluates deep data research over 10-K SEC filings, and the overall metric averages message-wise correctness (per-question factuality) and trajectory-wise correctness (multi-turn consistency).

TheAgentCompany (Xu et al., 2024) evaluates end-to-end task completion in a simulated enterprise environment, where each task requires a full containerized stack.

MARBLE (Zhu et al., 2025) evaluates open-ended research and buyer-seller negotiation under rubric-based scores within the MultiAgentBench framework.

AutomationBench (Shepard & Salimans, 2026) evaluates cross-application business-workflow orchestration through REST APIs, and the primary metric is accuracy across its 600 tasks.

OfficeQA Pro (Opsahl-Ong et al., 2026) evaluates enterprise-grade grounded numerical reasoning over the U.S. Treasury Bulletin corpus, with correctness scored under a 1% numerical tolerance.

B CASE STUDY OF PROTOCOL EVOLUTION

Figure 6 shows how the protocol evolves across rounds on the nine long-horizon and production-oriented benchmarks, which jointly cover repository-level coding, planning, retrieval, workflow orchestration, and open-ended research. On every benchmark the number of nodes remains small (≤ 4 across rounds), so evolution acts as a rewiring process over a protocol that stays small across very different task surfaces.

The three edit levels are unevenly used. Structural edits dominate, appearing in 22 of the 27 revisions in Figure 6, and the remaining five revise a predicate or the shared context without touching the graph. DevEval is the clearest case, where the second round moves unit-test injection into the shared rule and the third round bounds the review loop and records known error patterns, both without adding a role. Both rounds change how the protocol behaves while leaving the graph fixed, which a static topology has no place to record.

On the coding and planning benchmarks, evolution mainly redistributes verification responsibility. The SWE-bench Verified protocol first attaches a retry edge from the Verifier back to the Editor, then re-routes the Verifier back to the Locator, and finally splits the Verifier into structural and semantic components. DevEval gains a self-test node between the Coder and the Reviewer. DeepPlanning

grows from a monolithic solver into a Researcher-Solver-Verifier chain with an explicit constraint checklist.

On the retrieval and research benchmarks, evolution adds evidence-preserving roles. The DDR-Bench protocol fixes the Researcher-to-Reporter handoff to forward the raw evidence set, splits a single reporting phase into two, and installs a quantitative Auditor. OfficeQA Pro converts a monolithic solver into a Reader-Solver-Verifier chain over an external document store. MARBLE replaces the initial Reviewer-Brainstormer-Writer handoff with a compact Generator-Critic-Reviser pipeline and adds a feasibility-driven retry edge from the Critic back to the Generator.

Evolution also shrinks the graph when coordination is unnecessary. AutomationBench collapses a two-node Executor-Verifier pipeline into a single-node solver with progressively tightened internal phases, while HLE grows a confidence-routed subgraph that skips verification on high-confidence answers and adds a retry edge on refused ones. Section 5.2 locates the component measurement on which the accepted edits pay off.

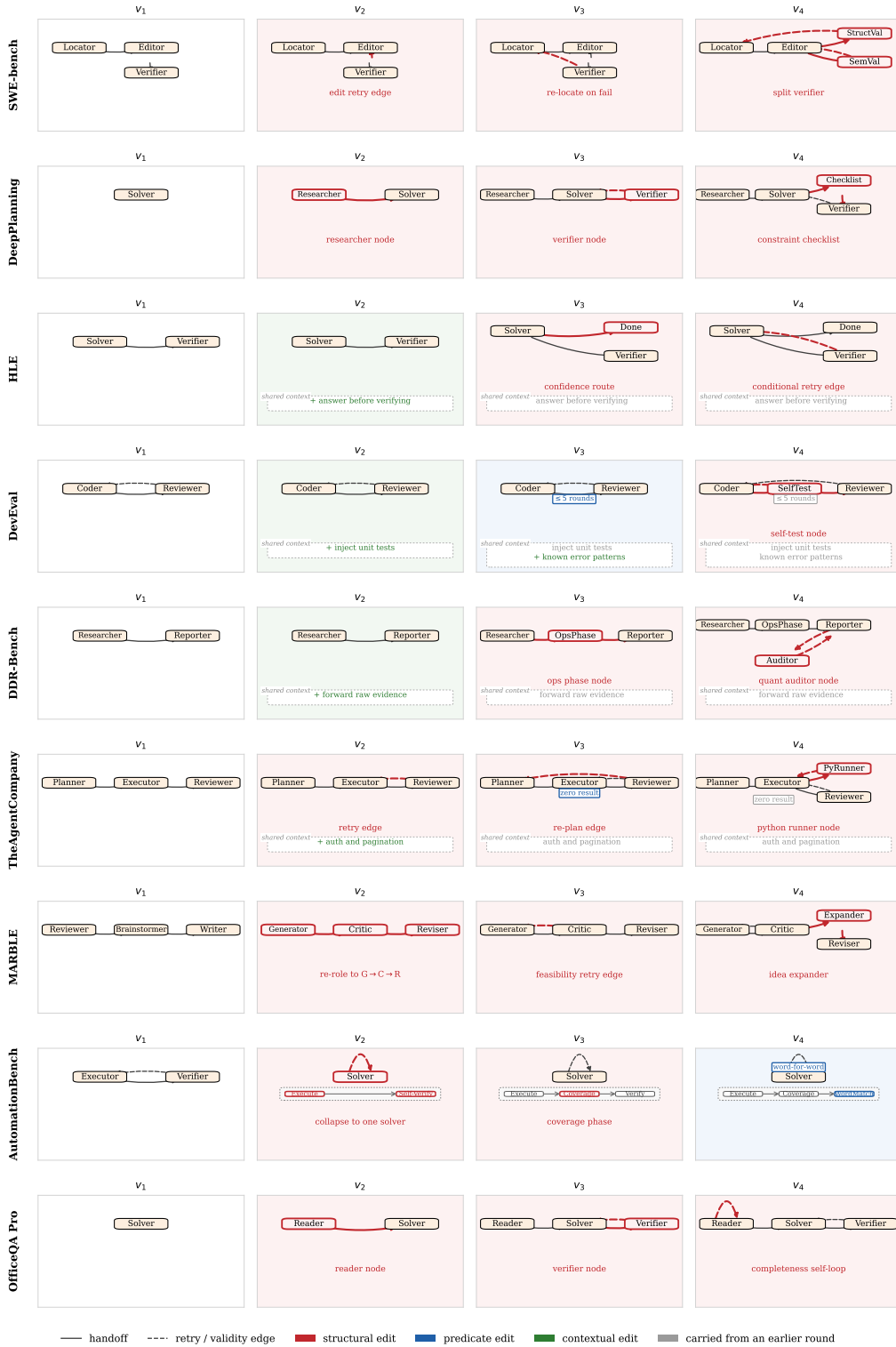


Figure 6: Protocol evolution across rounds on the nine long-horizon and production-oriented benchmarks, one per row. Solid ovals are agent roles, pills inside a dashed container are internal phases of a single-agent solver, solid edges are admissible handoffs, and dashed edges are validity or retry edges. Boxed text pinned to an element is a transition or completion predicate, and the lower container holds the shared rule and culture. Elements introduced by the current round carry the color of their edit level from Section 3.3 and earlier elements are gray, so a round that leaves the graph unchanged still shows what it revised.

Table 4: Governance boundary activity during the v_4 runs. Rejected is the percentage of proposed handoff events refused by $\Gamma(n, s)$, and Recovered is the percentage of runs whose outcome the refusal changed.

Benchmark	Rejected	Recovered
SWE-bench Verified	6.4	4.1
DevEval	4.9	3.7
DDR-Bench	3.8	2.9
TheAgentCompany	8.2	5.3

C ACTIVITY OF THE ADMISSIBLE HANDOFF SET

OpenAgency governs collaboration through the admissible handoff set $\Gamma(n, s)$, whose activity during execution determines how much of the collaboration structure is actually enforced. To verify that the boundary is active during execution, we measure how often $\Gamma(n, s)$ refuses a handoff an agent proposes. On the four benchmarks of Section 5.5 we log every proposed handoff during the v_4 runs, record the fraction $\Gamma(n, s)$ refuses, and count the runs whose outcome changes once the refusal redirects the handoff to an admissible target. As shown in Table 4, the boundary refuses 3.8% to 8.2% of proposed handoffs and changes the outcome of 2.9% to 5.3% of runs. Refusals are most frequent on TheAgentCompany, whose heterogeneous tool surface offers the most chances to address a collaborator that cannot serve the current state, and least frequent on the near-linear DDR-Bench pipeline. The boundary therefore stays active on all four benchmarks, ruling out the wrong collaborator at the points where it matters while leaving the common path unconstrained. These refused events are also the evidence the evolving agent consumes in Section 3.3.

Table 5: Primary metric across the four evolution rounds, each row in the benchmark’s native metric.

Benchmark	Single-agent	Multi-agent (init)	v_1	v_2	v_3	v_4
HumanEval	87.5	96.9	98.4	100.0	100.0	100.0
MBPP	73.5	73.5	74.3	80.5	80.5	80.9
GSM8K	97.2	97.4	98.0	98.0	98.0	98.0
MATH	82.3	86.0	87.0	87.0	87.0	87.0
HotpotQA	75.9	64.0	80.7	81.9	82.3	82.3
SWE-bench Verified	75.6	67.0	78.8	85.6	85.8	86.2
DeepPlanning	45.0	33.8	44.6	44.6	54.2	54.2
HLE (text-only)	18.4	5.6	33.7	35.8	35.9	37.7
DevEval	62.5	82.1	82.1	83.9	90.6	96.9
DDR-Bench	42.8	44.0	44.0	58.9	66.5	76.0
TheAgentCompany	46.3	41.5	46.3	48.8	51.2	53.7
MARBLE	78.4	50.0	73.4	91.7	95.0	98.9
AutomationBench	23.7	23.6	26.1	28.4	34.8	54.8
OfficeQA Pro	35.3	21.1	31.6	62.4	67.7	69.9

Table 6: Performance on HumanEval (Chen et al., 2021). The metric is pass@1.

Model / System	pass@1 (%)
Prior Ref.	92.2
OpenAgency (single-agent)	87.5
Multi-agent (init)	96.9
OpenAgency	100.0

D PER-BENCHMARK DETAILED RESULTS

This appendix reports the per-benchmark results summarized in Table 1, complementing the protocol-level traces of Appendix B. Table 5 lists the primary metric of all fourteen benchmarks across the four evolution rounds. Tables 6, 7, 8, 9, and 10 detail the general question answering benchmarks, Tables 11, 12, and 13 the long-horizon benchmarks, and Tables 14, 15, 16, 17, 18, and 19 the production-oriented benchmarks.

Table 7: Performance on MBPP (Austin et al., 2021). The metric is pass@1.

Model / System	pass@1 (%)
Prior Ref.	69.3
OpenAgency (single-agent)	73.5
Multi-agent (init)	73.5
OpenAgency	80.9

Table 8: Performance on GSM8K (Cobbe et al., 2021). The metric is exact-match accuracy.

Model / System	Exact match (%)
Prior Ref.	97.6
OpenAgency (single-agent)	97.2
Multi-agent (init)	97.4
OpenAgency	98.0

Table 9: Performance on MATH (Hendrycks et al., 2021). The metric is exact-match accuracy.

Model / System	Exact match (%)
Prior Ref.	83.3
OpenAgency (single-agent)	82.3
Multi-agent (init)	86.0
OpenAgency	87.0

Table 10: Performance on HotpotQA (Yang et al., 2018). The metric is token-level F1.

Model / System	F1 (%)
Prior Ref.	76.4
OpenAgency (single-agent)	75.9
Multi-agent (init)	64.0
OpenAgency	82.3

Table 11: Performance on SWE-bench Verified (Jimenez et al., 2024). FM is file match, and the first three rows are reported by that work.

Model / System	% Apply	FM	% Resolved
Claude-2	—	—	4.8
GPT-4	—	—	1.7
SWE-Llama-13B	—	—	3.9
Prior Ref.	72.4	42.8	15.4
OpenAgency (single-agent)	89.8	75.6	63.8
Multi-agent (init)	79.2	67.0	56.2
OpenAgency	100.0	86.2	74.4

Table 12: Performance on DeepPlanning (Zhang et al., 2026). Avg is the mean of travel and shopping case accuracy. External rows are reported by that work.

Model / System	Avg	Tra-CS	Tra-PS	Tra-Comp	Tra-Acc	Shop-M	Shop-Acc
Grok-4.1-Fast	3.0	39.6	3.0	29.6	0.0	50.1	5.9
GLM-4.7	7.1	38.9	7.1	30.7	0.0	61.2	14.2
Qwen-Plus	7.5	37.3	7.5	25.1	0.0	63.9	15.0
Seed-1.8	11.3	43.0	11.3	45.3	0.0	68.1	22.5
Kimi-K2-Thinking	12.1	45.2	32.5	38.9	0.0	65.8	24.2
DeepSeek-V3.2	21.6	47.4	35.0	41.2	0.7	78.8	42.5
Gemini-3-Pro	23.2	58.4	25.1	41.8	0.7	78.0	45.8
o3	24.9	76.5	55.6	66.1	11.3	76.9	38.5
Claude-Sonnet-4.5	25.5	65.2	58.4	61.8	7.6	80.0	43.3
Qwen3-Max	28.7	64.0	61.7	62.8	13.8	82.6	43.5
Gemini-3-Flash	28.8	67.1	57.7	62.4	5.9	80.6	51.7
GPT-5	31.6	78.7	65.9	72.3	18.9	80.4	44.2
Claude-Opus-4.5	33.9	79.3	70.9	75.1	22.7	80.0	45.0
GPT-5.2	44.6	88.5	83.3	85.8	35.0	84.8	54.2
Qwen-Agent tool-calling (Prior Ref.)	46.9	88.8	56.7	72.7	32.9	86.8	60.8
OpenAgency (single-agent)	45.0	73.9	67.5	70.7	35.0	85.2	55.0
Multi-agent (init)	33.8	44.5	38.3	41.4	26.7	76.2	40.8
OpenAgency	54.2	–	–	–	–	–	–

Table 13: Performance on the Humanity’s Last Exam text-only subset (Phan et al., 2025). External rows are reported by that work.

Model / System	Accuracy (%)
GPT-4o	2.7
Grok-2	3.2
Claude-Sonnet-3.5	4.3
Gemini-1.5-Pro	5.3
Gemini-2.0-Flash-Thinking	8.1
o1	8.1
o3-mini	18.0
Direct LLM	13.4
OpenAgency (single-agent)	18.4
Multi-agent (init)	5.6
OpenAgency	37.7

Table 14: Performance on DevEval (Li et al., 2024). UT is the unit-test pass rate and AT is the repository-level acceptance-test pass rate. The prior reference is a single coder node with no handoff, which is also the single-agent ablation.

Method	UT	AT
Direct LLM (Prior Ref.)	90.4	62.5
Multi-agent (init)	98.1	82.1
OpenAgency	99.1	96.9

Table 15: Performance on the DDR-Bench 10-K subset (Liu et al., 2026). Overall averages Message-Wise and Trajectory-Wise accuracy. External rows are reported by that work.

Model / System	Overall	Message-Wise	Trajectory-Wise
<i>Proprietary models</i>			
Gemini-2.5-Pro	20.1	24.5	15.6
Gemini-3-Flash	33.0	44.8	21.2
GPT-5.1	40.7	37.1	44.3
GPT-5-mini	42.0	46.8	37.1
GPT-5.2	43.0	44.9	41.1
Claude-Sonnet-4.5 (Prior Ref.)	69.1	77.6	60.6
<i>Open-source models</i>			
Llama-3.3-70B	6.5	9.9	3.0
Qwen2.5-72B	20.3	27.1	13.4
Qwen3-30B-A3B	28.4	42.4	14.4
MiniMax-M2	35.6	44.2	26.9
Qwen3-Next-80B-A3B	37.8	44.8	30.8
Kimi-K2	41.0	51.1	30.8
GLM-4.6	48.2	60.3	36.0
DeepSeek-V3.2	49.2	60.1	38.2
DDR reference agent	60.5	61.3	59.6
OpenAgency (single-agent)	42.8	27.3	58.2
Multi-agent (init)	44.0	–	–
OpenAgency	76.0	76.1	75.9

Table 16: Performance on TheAgentCompany (Xu et al., 2024). Score is the aggregated task score, and Steps is the average number of steps. External rows are reported by that work.

Model / System	Backbone	Success (%)	Score (%)	Steps (↓)
OpenHands	Gemini-2.5-Pro	30.3	39.1	33.5
OpenHands	Claude-Sonnet-3.7	26.3	34.7	40.9
OpenHands	Claude-Sonnet-3.5	24.0	34.4	29.2
OpenHands	Gemini-2.0-Flash	11.4	19.0	39.9
OpenHands	GPT-4o	8.6	16.7	28.7
OpenHands (Prior Ref.)	GPT-5.5	40.0	41.5	31.6
OpenAgency (single-agent)	GPT-5.5	46.7	46.3	27.4
Multi-agent (init)	GPT-5.5	–	41.5	–
OpenAgency	GPT-5.5	46.7	53.7	22.8

Table 17: Performance on MARBLE (Zhu et al., 2025). Overall averages Research and World scores. External rows are reported by that work.

Model / System	Backbone	Research (%)	World (%)	Overall (%)
Meta-Llama-3.1-8B	Llama-3.1-8B	80.9	72.8	76.9
Meta-Llama-3.1-70B	Llama-3.1-70B	80.8	72.1	76.5
Meta-Llama-3.3-70B	Llama-3.3-70B	80.0	73.2	76.6
GPT-3.5-turbo	GPT-3.5-turbo	70.2	71.7	71.0
GPT-4o-mini (Prior Ref.)	GPT-4o-mini	84.1	74.5	79.3
OpenAgency (single-agent)	GPT-5.5	66.8	90.0	78.4
Multi-agent (init)	GPT-5.5	–	–	50.0
OpenAgency	GPT-5.5	97.8	100.0	98.9

Table 18: Performance on AutomationBench (Shepard & Salimans, 2026). External rows are reported by that work.

Model / System	Acc (%)
GPT-6-Sol	33.2
GPT-6-Astra	41.4
Claude-Opus-5.5 (Prior Ref.)	42.5
OpenAgency (single-agent)	23.7
Multi-agent (init)	23.6
OpenAgency	54.8

Table 19: Performance on OfficeQA Pro (Opsahl-Ong et al., 2026). External rows are reported by that work.

Model / System	Corpus format	Acc@1% (%)
GPT-5-mini	Full PDF	30.8
GPT-5.4	Full PDF	36.1
Gemini-3.1-Pro	Full PDF	42.9
GPT-5.1	Full PDF	45.1
Claude-Opus-4.5	Full PDF	46.6
Claude-Opus-4.6 (Prior Ref.)	ai_parse_document	57.1
Direct LLM	Full PDF	26.3
OpenAgency (single-agent)	Full PDF	35.3
Multi-agent (init)	Full PDF	21.1
OpenAgency	Full PDF	69.9